# Jack Simon

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## Education

**Stony Brook University** Aug 2024 – present
*Undergraduate student*

- Coursework: ordinary differential equations, linear algebra.
- Directed Reading Program (Spring 2025): *Ordinary Differential Equations* by V. I. Arnold; final presentation.

## Research

**Huppert Lab** Aug 2026 – present
*Learned representations of fNIRS for functional brain mapping*

- Train CNNs on fNIRS signals projected into 10–20 scalp space to predict motor, language, social, and working-memory function.
- Study cross-subject prediction, cross-task generalization, and transfer (ImageNet → fMRI → fNIRS versus direct training), toward portable, at-home brain imaging.

**Nagan Lab (URECA, accepted Spring 2026)** Aug 2025 – present
*Conformational bias in RNA force fields*

- Compare torsion distributions across thousands of molecular dynamics frames to detect bias between force fields and solvent models, to inform parameter refinement.
- Debugging memory issues and porting the analysis to CUDA (in progress).

**Brookhaven National Laboratory, with Dr. Flynn** Spring 2026 – present
*Deep learning and optimization*

- Convolutional neural networks for image classification (MNIST, CIFAR).
- Planned for Summer 2027: open questions on rate-tracking smoothing for nonsmooth optimization.

## Experience

**AI Intern, Division of Information Technology (DoIT), Stony Brook University** Fall 2025

- Evaluated new LLMs, SLMs, and VLMs with structured SWOT analyses to recommend models for university adoption; presented findings to large audiences.

**Web Maintainer, Amber Tutorials (ambermd.org)** Aug 2025 – present

- Work with thousands of chemists to resolve issues and bring their tutorials and code into Amber.

## Presentations

- Delta-Goldstein, Clarke stationarity, smoothed gradient descent and optimization theory. Dr. Flynn's group, BNL, Nov 2026.
- Kepler's laws from the Newton–Euler laws and homothetic symmetry, Nov 2026.
- URECA poster: conformational bias in RNA force fields, Spring 2027 (upcoming).
- RNA torsions and biases; contributing to a website with version control. Nagan Lab.

## Teaching and Leadership

- Lead math tutor and outreach coordinator, Academic Success and Tutoring Center (math tutor since Jan 2025; lead since Fall 2026).
- Tutor and mentor, Stony Brook Athletics (Aug 2026 – present).
- Supervisor, Panera, Farmingdale (part time since Jan 2025; Employee of the Month Apr 2025; promoted Nov 2026).
- Volunteer: Island Harvest; crisis support through active listening.

## Skills and Projects

**Skills:** Python, PyTorch, CUDA, Git, cloud services; computer vision, data analysis, denoising.

**Projects:** NeuroBridge (fMRI-to-fNIRS model transfer, in development); ClearFile (document compliance app for Island Harvest, in development); Scope (math and algorithm visualization app, complete).